Customer Churn Analysis

Exploratory Data Analysis – Executive Summary

7,043 customers | 21 variables | Overall churn 26.54%

1. Overview

This exploratory data analysis focuses on understanding **customer churn** and identifying the customer characteristics, contract types, service choices, and payment methods associated with customers leaving the company.

The dataset contains **7,043 customers and 21 variables**, covering demographic information, tenure, subscribed services, contract details, payment methods, monthly charges, total charges, and churn status.

During data preparation, blank values in the `TotalCharges` column were replaced with **0** for customers with zero tenure, and the column was converted to a numerical format. After cleaning, the analysis found **no missing values** and **no duplicate customer IDs**, giving a clean basis for exploration.

2. Overall Churn Situation

The most important finding is that **26.54% of customers have churned**, while approximately **73.46% have stayed**. Roughly **1 out of every 4 customers** has left the company.

26.54%	73.46%	~1 in 4
Customers churned	Customers retained	Customers lost

Customer Status	Percentage
Stayed	73.46%
Churned	26.54%
Total	100%

Churn is a significant business issue, but it is **not evenly distributed**: certain customer segments churn at substantially higher levels than others.

3. Customer Demographics

Gender. Churn is present among both male and female customers, with **no major difference** between the two groups. Gender is therefore not one of the strongest variables for explaining churn compared with contract type, tenure, payment method, and additional services. Retention strategies based purely on gender would likely be less useful than strategies based on customer behavior and service relationships.

4. Senior Citizen Analysis

Approximately **16.21%** of customers are senior citizens and **83.79%** are non-senior. The churn proportion within the senior segment is considerably higher.

Customer Segment	Approx. Churn Rate
Non-Senior Citizens	23.6%
Senior Citizens	41.7%

Senior customers are an important segment for further investigation. Possible reasons to explore include service complexity, pricing, support requirements, contract preferences, or payment behavior.

> **Note:** The analysis does not establish that age itself causes churn; it identifies senior-citizen status as a segment with a higher observed churn rate.

5. Tenure and Customer Churn

Tenure is one of the clearest patterns in the analysis. Average customer tenure is approximately **32.4 months**, ranging from **0 to 72 months**. Customers with very short service histories, particularly those in the first one or two months, are more likely to churn, while customers with longer tenure show substantially more stable retention.

Short tenure → **higher churn**	Long tenure → **lower churn**

The **early customer lifecycle** appears to be a critical retention period. A customer who stays longer appears more likely to continue the relationship, so improving the first few months of the customer journey is an important area for investigation.

6. Contract Type – One of the Strongest Churn Indicators

Customers on **month-to-month contracts churn at a much higher rate** than customers with one-year or two-year contracts.

Contract Type	Approx. Churn Rate
Month-to-Month	42.7%
One Year	11.3%
Two Year	2.8%

This makes contract type one of the most important segmentation variables identified.

> **Interpret with care:** this is an association, not proof of causation. Customers who choose longer contracts may already differ from month-to-month customers in other ways.

7. Service Usage and Churn

The analysis examined the relationship between churn and: Phone Service, Multiple Lines, Internet Service, Online Security, Online Backup, Device Protection, Tech Support, Streaming TV, and Streaming Movies.

Customers using additional **security, backup, device-protection, and technical-support services** generally show lower churn than customers who do not.

Service	Customer Group	Approx. Churn Rate
Online Security	No	41.8%
Online Security	Yes	14.6%
Online Backup	No	39.9%
Online Backup	Yes	21.6%
Device Protection	No	39.1%
Device Protection	Yes	22.5%
Tech Support	No	41.6%
Tech Support	Yes	15.2%

The gaps are most noticeable for **Online Security** (41.8% vs. 14.6%) and **Tech Support** (41.6% vs. 15.2%). Customers more engaged with support and protection services tend to show stronger retention.

> **Note:** These results do not prove that adding a service will prevent churn. Service adoption may also relate to contract type, tenure, or customer needs.

8. Payment Method and Churn

Customers paying by **Electronic Check** show a substantially higher churn rate than customers using other payment methods, almost half of that segment has churned.

Payment Method	Approx. Churn Rate
Electronic Check	45.3%
Mailed Check	19.1%
Bank Transfer – Automatic	16.7%
Credit Card – Automatic	15.2%

Payment behavior could help identify customers who need additional retention attention. Further analysis is needed to determine whether the payment method itself matters or is linked to other customer characteristics.

9. Key Business Insights

Higher observed churn

- **Month-to-month** customers – ~42.7%
- **Electronic Check** customers – ~45.3%
- **Senior Citizens** – ~41.7%
- **Very short-tenure** customers (first few months)
- Customers **without** Online Security / Tech Support

Lower observed churn

- **Two-year** contracts – ~2.8%
- **One-year** contracts – ~11.3%
- Customers on **automatic payment** methods
- Customers with **longer tenure**
- Customers using security and tech-support services

Churn is strongly associated with a combination of **tenure, contract commitment, payment behavior, and service engagement**.

10. Overall Business Interpretation

Churn is concentrated among customers with a **shorter relationship**, **less contractual commitment**, certain **payment behaviors**, and **lower adoption** of support or protection services. The most prominent patterns:

1	**26.54%** overall churn
2	Short-tenure customers show **higher churn**
3	Month-to-month customers show **~42.7%** churn
4	Electronic Check customers show **~45.3%** churn
5	Senior Citizens show **~41.7%** churn
6	Customers without security/support services generally show **higher churn**

Churn analysis should not focus on a single variable; retention relates to a combination of **customer lifecycle, contract structure, payment behavior, and service engagement**.

11. Recommended Areas for Business Attention

1. Focus on the early customer lifecycle

Short-tenure customers show higher churn. Investigate onboarding, service quality, pricing, customer support, and early engagement in the first few months.

2. Understand month-to-month customers

With ~42.7% churn, this segment merits deeper analysis: why customers stay on flexible contracts, and whether different retention offers or service experiences change their behavior.

3. Investigate Electronic Check customers

At ~45.3% churn, well above automatic payment users. Examine whether the link runs through contract type, tenure, monthly charges, or other characteristics.

4. Increase engagement with support and security services

Lower churn is observed with Online Security, Online Backup, Device Protection, and Tech Support. Investigate whether greater awareness or adoption is associated with stronger retention.

5. Examine senior-citizen customer experience

With ~41.7% churn, avoid assuming the reason. Examine contract type, tenure, payment method, service usage, and support interactions within this segment.

12. Final Executive Takeaway

The analysis of **7,043 customers** shows an overall churn rate of **26.54%**, meaning more than one-quarter of the customer base has left.

Churn is concentrated among **month-to-month customers, Electronic Check users, senior citizens, and short-tenure customers**. Customers with longer contracts, longer tenure, automatic payment methods, and additional security or technical-support services generally show lower observed churn.

The strongest numerical differences appear in **contract type** (42.7% month-to-month vs. 11.3% one-year vs. 2.8% two-year) and **payment method** (45.3% Electronic Check vs. roughly 15–17% for automatic payments). Customers using Online Security, Online Backup, Device Protection, and Tech Support also show lower churn.

> **Next step:** Build a **customer churn prediction model** that combines these variables into a single risk score, moving from descriptive analysis toward proactive customer-retention decisions.